The Pirate Cove Puzzle

Whodunnit? · Basics & Reasoning Name: _____

Aboard the sailing ship *Sea Sparrow*, the crew was ready to set sail. The Captain reached for the magic **Golden Compass** that always points to treasure, but its hook was bare!

Six crew members were on the ship and shore, each in a different place, and there are six ways a compass could be smuggled away. You are the ship's detective: find **who** took it, **where** they were, and **how** they got it off the ship.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ First Mate Morgan Ship's Officer	☐ Crow's Nest	☐ Buried in the sand
☐ Salty Sue Ship's Cook	☐ Captain's Cabin	☐ Hidden in an apple barrel
☐ Navigator Dax Map Reader	☐ Galley	☐ Strapped to the parrot
☐ Polly Finch Parrot Trainer	☐ Cargo Hold	☐ Wrapped in a sail
☐ Deckhand Finn Sailor	☐ Plank Deck	☐ Rowed off in a dinghy
☐ Cora Blake Captain's Cousin	☐ Treasure Cave	☐ Sealed in a message bottle

The culprit is _____ in _____ using _____

Show Your Work

The Pirate Cove Puzzle · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Pirate Cove Puzzle

M is the midpoint of AB. AM = 3x + 2 and MB = 5x − 6. Find the length of AB.

A. 52	Every message bottle on the ship was held up to the light. None had a compass inside.
B. 28	First Mate Morgan was scanning the horizon from the Crow’s Nest. Two sailors waved to him.
C. 4	First Mate Morgan was mapping the tide pools in the Treasure Cave. His boots are still soggy.
D. 14	Navigator Dax was checking the stars from the Crow’s Nest. The lookout swears he never climbed down.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Pirate Cove Puzzle

Two angles are complementary. One measures x° and the other measures (2x + 15)°. Find the larger angle.

A. 65°	The beach was combed with a metal detector. Nothing buried in the sand.
B. 85°	The message bottles on the shelf are all corked with dusty wax. None was opened or resealed.
C. 125°	Salty Sue was hunting for crab shells in the Treasure Cave. Her full basket proves it.
D. 25°	Navigator Dax was polishing the Captain’s telescope in the Captain’s Cabin all evening.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Pirate Cove Puzzle

Two angles are supplementary and measure (3x + 10)° and (5x − 6)°. Find the smaller angle.

A. 22°	Polly Finch was teaching the parrot to squawk in the Treasure Cave. The echo was deafening.
B. 79°	The bottle washed up on shore holds only a note about lunch. No compass was sealed inside.
C. 104°	Navigator Dax was tasting soup in the Galley. The cook gave him two bowls.
D. 76°	Salty Sue was stirring stew in the Galley. Half the crew came by for seconds.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Pirate Cove Puzzle

Ray BD bisects ∠ABC. m∠ABD = (3x + 5)° and m∠DBC = (5x − 19)°. Find m∠ABC.

A. 12°	Deckhand Finn was lighting lanterns in the Treasure Cave. The wax drips are still warm.
B. 164°	The neck of the message bottle is too narrow for the Golden Compass to fit through.
C. 82°	The apple barrel was emptied: only apples and one very surprised mouse.
D. 41°	Navigator Dax was counting barrels in the Cargo Hold with the quartermaster.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Pirate Cove Puzzle

Find the distance between (1, 2) and (4, 6).

A. 7	Navigator Dax was swabbing the Plank Deck beside Deckhand Finn. The mop buckets prove it.
B. 10	The harbor master collected every floating bottle at dawn. All were empty.
C. 5	Polly Finch was feeding the parrot in the Cargo Hold with a crate of crackers.
D. 25	Cora Blake was sketching stalactites in the Treasure Cave. Her charcoal smudges are everywhere.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Pirate Cove Puzzle

Find the midpoint of the segment with endpoints (−2, 6) and (8, −2).

A. (3, 2)	The parrot can only carry crackers. No compass strapped to it.
B. (6, 4)	Navigator Dax was waving flags from the Crow’s Nest. A passing ship waved back.
C. (5, −4)	First Mate Morgan was chasing a crab around the Treasure Cave. The crab won.
D. (3, 4)	The ink on the bottle’s note is dry and old. No one sealed anything in it last night.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Pirate Cove Puzzle

Two vertical angles measure (4x − 7)° and (3x + 14)°. Find the measure of each angle.

A. 77°	Deckhand Finn was scrubbing the Plank Deck. The soap-bubble trail proves it.
B. 154°	Salty Sue was gathering seaweed in the Treasure Cave. Her bucket was full when she came out.
C. 21°	Navigator Dax was signing logbooks in the Captain’s Cabin. The Captain stamped each page.
D. 103°	The cork in every message bottle is cracked and loose. None could stay sealed at sea.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Pirate Cove Puzzle

Points A, B, C are collinear with B between A and C. AB = 12, BC = 2x + 1 and AC = 5x − 2. Find BC.

A. 23	Polly Finch was whistling sea shanties in the Treasure Cave. The echoes gave her away.
B. 13	The message bottles are tied in a net below deck. The knots were never touched.
C. 5	Navigator Dax was peeling potatoes in the Galley. The pile grew taller by the minute.
D. 11	The sails were unrolled and shaken out. Nothing was wrapped inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Pirate Cove Puzzle

Two angles form a linear pair and measure (2x + 20)° and (4x − 2)°. Find the larger angle.

A. 74°	Navigator Dax was stacking crates in the Cargo Hold. The crew counted every one.
B. 106°	Cora Blake was reading maps with the Captain in the Captain’s Cabin.
C. 46°	The sea was too rough to float any bottle last night. Not one left the ship.
D. 27°	Deckhand Finn was digging for shells in the Treasure Cave. Sand still fills his boots.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Pirate Cove Puzzle

How many different line segments can be named using any two of 6 collinear points as endpoints?

A. 30	Navigator Dax was fishing off the Plank Deck. The ship’s cat watched him all night.
B. 36	The last message bottle was found full of seawater. Nothing dry could have been sealed in it.
C. 15	The dinghy was tied up and dry all night. No one rowed away.
D. 21	Cora Blake was resting on a rock in the Treasure Cave, waiting for the tide to turn.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.